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RespiScan: An AI-Powered Software using CNN-Based AI Models in Bacterial Pneumonia in Chest X-rays

CC303-Methods of Research

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CHAPTER 1

Introduction

1.1 Background of the Study

Bacterial pneumonia remains a leading cause of mortality worldwide, with Streptococcus pneumoniae as the primary pathogen responsible for the majority of cases. According to the Global Burden of Disease report cited in recent analyses, pneumonia contributed to over 2 million deaths in 2021, with bacterial forms exacerbating antimicrobial resistance challenges. Traditional diagnosis, according to Arizmendi, Pinto, Arboleda, and González (2025), relies on chest X-rays, but radiologist shortages and interpretation variability necessitate AI assistance; for instance, CNN models have achieved up to 93.66% accuracy in distinguishing bacterial from viral pneumonia. This thesis proposes a CNN-based object detection system tailored for bacterial pneumonia to enable rapid, precise localization on X-rays, aiding doctors in resource-limited settings like the Philippines.

In current medical practice, chest X-rays serve as the first-line imaging modality for respiratory infections due to their wide availability, affordability, and speed. However, interpretation accuracy can vary significantly across clinicians, particularly in under-resourced healthcare environments where trained radiologists are scarce. This limitation can delay diagnosis and affect treatment outcomes, leading to prolonged hospitalization or higher mortality rates. Automated diagnostic solutions powered by artificial intelligence (AI) can potentially standardize interpretation, reduce diagnostic delays, and serve as a triage tool for prioritizing high-risk patients.

Deep learning algorithms, particularly Convolutional Neural Networks (CNNs), have revolutionized image analysis in recent years due to their ability to learn complex features directly from data. As demonstrated by Yanar, Hardalaç, and Ayturan (2025), CNN-based models can effectively detect pneumonia-related abnormalities in chest radiographs with high precision and recall. Rahman et al. (2021) further showed that incorporating preprocessing and image enhancement techniques can significantly improve diagnostic accuracy, highlighting how image quality optimization can amplify model performance. Similarly, early studies by Stephen et al. (2019) established the feasibility of deploying deep learning models in real-world clinical settings for pneumonia detection, underscoring their role as support tools for healthcare professionals.

A key advancement in the clinical deployment of AI systems is the integration of explainable AI techniques that enhance trust and transparency. Visualization methods such as Gradient-weighted Class Activation Mapping (Grad-CAM) and saliency maps can provide heatmap annotations of pneumonia-affected regions, allowing medical professionals to interpret the model's focus areas. As emphasized by Müller, Soto-Rey, and Kramer (2022), model evaluation in medical imaging must prioritize reliability and reproducibility, employing metrics such as sensitivity, specificity, F1 score, and AUC-ROC to ensure dependable clinical decisions.

Given these factors, the proposed system, RespiScan, aims to bridge the diagnostic gap by utilizing a CNN-based approach capable of detecting and localizing bacterial pneumonia in chest X-rays. Through annotated heatmaps and interpretable outputs, the system is envisioned to support pulmonologists and general practitioners in making faster and more reliable diagnostic decisions. Ultimately, the goal is to contribute to early detection, timely treatment, and reduction in the overall burden of pneumonia, particularly in regions with limited medical manpower and diagnostic infrastructure.

1.2 Project Objectives

To design and develop an AI-powered software capable of detecting early to late stages of Bacterial Pneumonia allowing clinicians better medical diagnostics regarding Bacterial Pneumonia from X-ray images.

Specific Objectives:

1. To design and develop a computer vision model capable of detecting Bacterial Pneumonia related anomalies from X-ray images.
2. To create a software that visualizes findings through annotated heatmaps to aid pulmonologists regarding clinical decision making.
3. To test and evaluate the performance of the software based on model's sensitivity (precision and recall), specificity, F1-score, Intersection-over-Union (IoU), Area under ROC Curve (AUC-ROC) in determining records that were diagnosed with Bacterial Pneumonia from the ground truth by medical professionals.

4. To evaluate the level of acceptability of the system based on the criteria of ISO 25010.

1.3 Significance of the Study

For Healthcare Professionals and Radiologists

- Enhancing Diagnostic Speed: Automating the initial screening of X-rays to allow for rapid triage of high-risk bacterial pneumonia cases.
- Reducing Subjectivity: Standardizing the interpretation of chest radiographs, thereby minimizing human error and variability between different levels of clinical expertise.
- Providing Visual Evidence: Using Grad-CAM heatmaps to highlight pathological regions (such as consolidations or infiltrates), which fosters trust and allows doctors to validate AI findings with their own clinical observations.

For Patients and Public Health

- Improved Patient Outcomes: Facilitating early intervention, which reduces the risk of severe complications, prolonged hospitalization, and death.
- Combating Antimicrobial Resistance (AMR): By accurately distinguishing bacterial pneumonia from other respiratory conditions, the system helps clinicians prescribe appropriate antibiotics, preventing the misuse of medication that fuels AMR.

- Health Equity: Providing high-level diagnostic capabilities to rural and under-resourced hospitals in the Philippines, ensuring that quality care is not limited to urban medical centers.

For the Field of Medical AI and Software Engineering

- Advancing Explainable AI (XAI): Demonstrating how complex "black-box" CNN models can be made transparent and clinically relevant through localization techniques.
- Setting Quality Benchmarks: Utilizing the ISO 25010 framework to evaluate the software ensures that the system is not only accurate but also reliable, maintainable, and functionally suitable for real-world medical environments.
- Establishing Evaluation Standards: Using rigorous metrics such as AUC-ROC, F1 score, and specificity provides a blueprint for future researchers to validate deep learning models in infectious disease diagnostics.

For the Philippine Healthcare System

RespiScan aligns with the National AI Strategy Roadmap of the Philippines by leveraging technology to address local health disparities. It serves as a scalable model for how AI can be integrated into the national health infrastructure to augment the capabilities of general practitioners and community health workers.

1.4 Scope and Limitations of the Study

- **Data Source:** The study will utilize publicly available, anonymized datasets of chest X-rays (such as the ChestX-ray14 or the Kermany/Mendeley dataset) specifically labeled for Bacterial Pneumonia.

- **Model Architecture:** The software will focus on Convolutional Neural Networks (CNNs), specifically architectures optimized for object detection and classification (e.g., ResNet, Xception, or YOLO variants).
- **Target Pathology:** The primary focus is the detection and localization of Bacterial Pneumonia, distinguishing it from normal lung conditions and viral pneumonia.
- **Software Features:**
    - Image preprocessing (noise reduction and contrast enhancement).
    - Classification of X-ray images into "Bacterial Pneumonia" or "Normal/Other."
    - Generation of Grad-CAM heatmaps to highlight the specific lung regions where the model detects anomalies.
- **Technical Evaluation:** Performance will be measured using standard metrics: Sensitivity (Recall), Precision, Specificity, F1 Score, and AUC-ROC.

Delimitations:

- **Real-time Clinical Integration:** The study focuses on the development and testing of the software; it will not involve real-time integration into Hospital Information Systems (HIS) or Electronic Health Records (EHR) during this phase.
- **Other Respiratory Diseases:** While the model may encounter other conditions, it is not specifically designed to diagnose tuberculosis, lung cancer, or COPD, unless they mimic bacterial pneumonia patterns.
- **Hardware Development:** The project is strictly software-based and will not involve the creation of new X-ray hardware or physical sensors.

- **Patient Demographics:** The study will not account for variables such as age, gender, or pre-existing comorbidities unless that data is explicitly provided within the training datasets.
- **Clinical Treatment**: The software is a diagnostic aid and will not provide treatment recommendations or antibiotic prescriptions.

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CHAPTER 2

Review of Related Literatures

2.1 Review of Related Literatures and Studies

Clinical Foundations of Bacterial Pneumonia

Bacterial pneumonia is characterized by an acute infection of the lung parenchyma in which fibrin, inflammatory cells, and exudate fill the alveolar air gaps. Clinically, this inflammation sets off a host immunological reaction that shows up as pleuritic chest discomfort, fever, and a productive cough with purulent sputum. Severe consequences include necrotizing pneumonia, empyema, and potentially fatal sepsis can develop from an untreated infection (Sattar et al., 2024).

From a diagnostic perspective, bacterial pneumonia is frequently classified according to its anatomical distribution, such as lobar pneumonia, which affects a particular lung segment (Sattar et al., 2024). For physicians, this lobar consolidation serves as the main visual cue. However, the disease's resemblance to "atypical" pneumonia—which might not show up on conventional Gram stains—highlights the need for reliable imaging instruments that can identify a range of clinical manifestations (Sattar et al., 2024).

Radiological Patterns and Imaging Challenges

Because of their accessibility and capacity to show lung disease, chest X-rays (CXR) are essential to the pneumonia diagnostic paradigm (Cook et al., 2022). Air bronchograms, which show air-filled airways against opaque, fluid-filled lung tissue, and homogeneous consolidations are typical results (Cook et al., 2022). Because these patterns frequently overlap with inflammatory diseases, vasculitis, or even lung cancer, imaging alone is rarely able to reach a definitive decision, even while these patterns offer hints for narrowing down a diagnosis (Cook et al., 2022).

This overlap creates a significant challenge for human radiologists, as the subjective nature of manual reading can lead to misdiagnosis or delayed treatment (Cook et al., 2022). Consequently, there is a growing clinical requirement for "contextualized" diagnostic tools that integrate imaging with laboratory data (Cook et al., 2022). For software like RespiScan, this highlights the importance of not just detecting opacities but providing high-precision feature extraction to help clinicians distinguish pneumonia from look-alike diseases (Cook et al., 2022).

Artificial Intelligence and Medical Imaging

The growing application of artificial intelligence in medical imaging has significantly advanced the diagnosis of bacterial pneumonia through chest X-ray (CXR) analysis, offering a promising solution to radiologist shortages and the need for earlier, more accurate detection.

Several studies have demonstrated strong diagnostic performance using convolutional neural network (CNN)-based architectures. Ippolito et al. (2023) developed an AI-assisted tool for emergency differential diagnosis, achieving 92% sensitivity and 89% specificity in distinguishing bacterial from viral pneumonia using CNNs on CXRs, underscoring its reliability in time-critical clinical environments. Kadali et al. (2023) similarly applied ResNet-50 on augmented datasets, attaining 95% precision and validating the effectiveness of transfer learning in identifying bacterial infection patterns. Building on this, Aydin et al. (2025) proposed a hybrid

CNN-ResNet model that achieved 96.5% accuracy by leveraging feature extraction from lobar consolidations characteristic of bacterial pneumonia.

Attention mechanisms have further elevated classification performance. Potharaju et al. (2025) integrated Convolutional Block Attention Module (CBAM) and Squeeze-and-Excitation (SE) attention into CNN architectures, achieving 98.6% accuracy by directing the model's focus toward key lung opacities. Li (2024) complemented this approach with an attention-enhanced ResNet using focal loss, reaching 98% accuracy while effectively addressing class imbalance by prioritizing pneumonia-critical regions.

Object detection and localization have also been a focus of recent work. Al Foysal and Sultana (2025) compared CNN architectures including ResNet-50 and VGG16, finding ResNet superior with 94.2% accuracy in pneumonia localization — an outcome particularly relevant for resource-limited clinical settings. Munna et al. (2024) evaluated multiple YOLO variants, with YOLOv8 achieving a mean Average Precision (mAP) of 92% for pneumonia bounding box detection, establishing it as a strong candidate for localization tasks in systems like RespiScan.

Image quality and preprocessing have received attention as well. Xu and Zhang (2026) employed a Double SGAN framework for CXR enhancement, improving CNN-based detection to a 97% F1-score and aiding the identification of subtle early-stage bacterial features.

Explainability and clinical trustworthiness have emerged as critical considerations alongside raw performance. Slimi et al. (2025) introduced an attention-guided spiking neural network for pneumonia detection, achieving 96.48% accuracy and 99.82% AUC while producing interpretable outputs aligned with clinically relevant regions. Rabbah et al. (2025) similarly combined CNNs with integrated gradients, yielding 97% accuracy alongside explainable heatmaps to support clinical confidence. Hedhoud et al. (2023) proposed a CNN-XGBoost hybrid for bacterial and viral differentiation, reaching 87% accuracy and 89% specificity while outperforming baseline models.

On the methodological front, Laos et al. (2025) conducted a systematic review of deep learning approaches for early pneumonia detection, reporting that CNNs augmented with data augmentation techniques achieve up to 98.94% accuracy, providing a useful framework for principled model selection.

Collectively, these works establish a strong foundation for RespiScan's development, affirming CNNs enhanced with attention mechanisms and explainable AI (XAI) as the state-of-the-art in moving from simple classification toward interpretable, clinically applicable pneumonia detection.

Global Radiology Workforce

The need for diagnostic imaging has been at a steady increase globally due to the aging population, cancer now identified as a chronic disease, and the significant increase in the complexity of each study, making interpretation time-consuming (Matsumoto, 2024). Despite rising demand, the number of radiologists has not increased at a satisfactory rate, even with increased financial incentives, a negative work-life balance, and a shortage of continuing professional development contributing to a decline in the workforce (Achour et al., 2025).

Achour et al. (2025) talks about additional reports that show other countries, including Poland, South Africa, Australia, and Korea, reporting a similar challenge. They also state that it is even worse in low to middle-income countries due to limited staff and imaging resources.

Radiology is crucial in modern day medical diagnostics. However, persistent shortage in radiologists will severely affect the medical industry where image diagnostics are needed the most. As years passes by, the demand for medical imaging continues to rise, but the radiology workforce grows slower, and eventually will be outpaced by its demand based on the projection made by the The Radiological Society of North America (RSNA) (Chopra, 2026). This increasing imaging demand also put radiologists under pressure leading to burnout. Chopra cited a study conducted by Lee et al. in 2025 about radiology workloads, aside from imaging diagnostics demands, administrative workloads and compliances lead to backlogs piling up and delayed follow-ups, treatment, and diagnostics. This was also proved in an article from PhillyVoice by Bond in 2026, where researchers who studied the wait times of 2.6 million of CTs, MRIs, ultrasounds and other scans conducted between 2014 and 2023, said that waiting times for medical imaging results doubled while suffering from radiologist shortage. This proves that persistent shortage puts healthcare quality and patient treatment at risk.

A study by Alsalah (2025) shows that even in countries that possess a relatively high number of radiologists, the workforce is rarely distributed properly. This can be traced by matching the distribution of training centers, specialist hospitals, and advanced imaging equipment to the number of radiologists in a specific location. For example, the geographic distribution of the radiologist workforce in Saudi Arabia shows that 67% of their radiologist workforce is concentrated in just three large population provinces (Riyadh, Mecca, and the Eastern Region).

The cost of these factors such as workloads, job demands, work-life integration, social support, etc. is severe occupational burnout (Parikh and Lexa, 2026). According to Ruff (2025), over the last two decades, there has been a gradual shift in the balance of new and retiring radiologists. The workload dedicated to training residents have decreased by 25%, while the overall workload for radiologists has skyrocketed to 80%. Driven by rising workloads, and the neet to maintain productivity, the workforce face emotional exhaustion, social isolation, and extended work-hours, which all contribute to the shortage crisis (Parikh and Lexa, 2026).

Struggling Philippine Radiology Industry in Recent Years

The Philippines currently suffers from national shortage of medical professionals which becomes a more concerning factor in aligning with the "8-Point Action Agenda as the Medium-Term Strategy of the Health Sector" that is currently adopted as of the release of Administrative Order No. 2023-0015 by the Department of Health. The government should ensure that every Filipino should be served with quality, affordable, accessible healthcare at all times. The shortage will hinder the health sector from realizing their goals which are better health outcomes, stronger health systems, and access to all levels of care.

A market analysis made by Aarti Patel about the radiology industry of the Philippines was published by a healthcare-focused research firm Insights10 from 2024. Patel pointed out factors contributing to the epidemic of chronic diseases, market growth drivers which makes the healthcare industry to grow, and market restraints such as high cost and poor healthcare facilities and the shortage of radiologists with a number of 2,200 active radiologists. Those restraints, especially the shortage, hurt both the people and the radiology market as the demand for radiology services exceeds the current available manpower considering that those 2,200 radiologists will serve 110 million Filipinos. This means there will be only 2 radiologists per 100,000 Filipinos, compared to the whole of Europe that has 13 per 100,000 (Patnaik, n.d) and developing countries like the United Kingdom and Singapore that have 9.9 per 100,000 (The Royal College of Radiologists, 2023) and 7.6 per 100,000 respectively (Rappler, 2024).

According to the reports made by the Professional Regulation Commission about Radiologic Technologist Licensure Examination from 2021 to 2025, there is an increasing number of board passers from 786 passers in 2021 to 2,629 passers in 2025. This implies that many people have taken their ways to contribute to the struggling industry of radiology. However, this seemingly increasing number of passers will still not suffice the national deficit of nearly 19,000 radiologic technologists based on the Private Sector Advisory Council's report on a health human resource crisis in 2026.

As of January 14, 2025 based on a document uploaded by the Professional Regulation Commission in the eFOI portal, there are 28,127 registered radiologic technologists in the Philippines. However, only 19,732 of them are active professionals which was only 70 percent of registered radiographers based on a press release published by the University of the Philippines - Manila on the 30th of April, 2026. It was also mentioned in the press that the university decided to contribute to the national shortage in radiologic technologists by opening the Bachelor of

Science in Radiologic Technology in UP Manila. This was a response of the University of the Philippines - Manila for the national shortage in human health resources, specifically for the radiology field.

Artificial Intelligence as a Mitigation Strategy

A critical development in X-ray technology is being employed, shifting the traditional legacy X-ray machine into a smaller, more efficient portable X-ray device. Healthcare providers are able to do their jobs quickly, provide diagnoses, and communicate the findings to other workers connected to the case (Ruff, 2025). While these changes may bring short-term relief for our workforce, a bigger technological change is needed to handle the problem on a global scale (Matsumoto, 2024).

According to Ruff (2025), the development of Artificial Intelligence (AI) and Machine Learning (ML) offers an increase in efficiency by automating simple and repetitive tasks. AI can act as a virtual assistant that streamlines the workflow, further reducing the burden on medical professionals.

Despite the premise of AI being the solution to ending the problem, recent literature warns that it's not a perfect solution and that heavy reliance on AI poses its own set of problems. It operates like a "Black Box," meaning radiologists do not fully understand the complex nature of the deep learning layers (Parikh & Lexa, 2026; Achour et al., 2025).

Developing AI models that provide accuracy comfortable enough to reduce doubt and overcome clinical hesitation presents a paradoxical challenge. Creating an effective AI tool relies on and requires a heavy amount of high-quality annotated data. However, annotating medical images used for training requires a large amount of time and clinical expertise. This creates a bottleneck, as we are already experiencing a shortage of radiologists (Gabriel et al., 2025).

Evaluation Metrics for AI Medical Imaging

Technology evolves rapidly enough for the people to witness machines or computers exhibit skills and abilities on par to humans. In the AI field, machine learning enables computers to possess human-like intelligence, such as sound reasoning, conversing ideas, recognizing patterns, detecting objects through a camera, or predicting outcomes based on calculations. When evaluating a computer vision model for medical imaging, it is crucial to make the model do precise readings and exhibit accurate performance. It should be expected that the model will perform comparable to the professionals in a specific domain or field it was made for. Therefore, determining proper metrics for evaluating models will be essential in fine-tuning as they aid in analyzing model performance and distinguishing areas that need attention for improvement.

A study conducted by Schlosser et al. published in March 2026 provides a comprehensive overview about different evaluation and performance metrics that serves as a guideline for appropriate assessment and validation for machine learning models. Models are typically measured by its precision, recall, and accuracy. Precision metric is used for measuring the accuracy of positive predictions. This metric tells how many predicted positives are actual positives and useful for minimizing false positives. Recall, sensitivity, or true positive rate complements precision as it shows if the model correctly identified actual positives. This metric ensures minimal false negatives, but ignores true negatives and false positives. Specificity, selectivity, or true negative rate metric emphasizes the relation of predicted true negatives to false positives. High specificity suggests that the model has a low chance of predicting false positives, leading to less number of further check ups and treatment. However, it is advised to complement results of specificity with recall, as low recall indicates that the model fails to predict actual positives and negatives. Accuracy provides the overall performance of the model can be misleading, especially to unbalanced datasets, while balanced accuracy is ideally used for unbalanced datasets since it considers sensitivity and specificity of the model.

Furthermore, F-score is useful when determining whether the model performs great in predicting all actual positives by balancing precision and recall, and false positives are more devastating, which makes it an essential metric for medical imaging. F-score has many variations for specific applications and trade-offs such as F0 if precision is maximized, F0.5 for increased importance of precision, F1 for balanced precision and recall, F2, if recall is more important than precision. (Guarav, 2023). High F1-score indicates high precision and high recall simultaneously. The Area Under the Curve and Receiver Operating Characteristic curve (AUC-ROC curve) are primarily used for classification models, especially for imbalanced datasets and binary classes. This metric measures the discriminatory ability of the model, how well the model separates objects between classes (e.g., background pixels from actual object pixels, infected lung from healthy lung). High AUC value indicates that the model correctly classifies objects rather than randomly guessing. In medical imaging, models should always classify and predict actual positives and negatives rather than giving false positives or readings.

Evaluating computer vision models is also gauging its localization ability, or locating the area of interest within a frame or an image with bounding boxes. Based on a study conducted by Müller et al. in 2022, sensitivity and specificity are also used in object detection. However, these two metrics are only suited for assessing the model functionality which is primarily to classify, and not its performance. Therefore, it is crucial to evaluate how it annotates or segments different sections it identifies as the objects of interest in an image. The Intersection over Union (IoU) metric, also known as Jaccard Index, is primarily used to determine the model's accuracy of predicted annotation or segmentation to the ground truth, which is the pixels that was initially regarded as the object. This metric also includes precision, thus ensuring minimal false positives which is common especially to highly imbalanced classes which are common in medical imaging (Matzunaka & Mazayuki, 2026). Another metric, the F1-score, or also known as Dice Similarity Coefficient (DSC), is also used for evaluating model localization. It assesses similarly to IoU, however is less stricter to annotation and segmentation overlaps and underlaps than IoU. The only difference between F1-score and DSC is the context usage as the latter is primarily used for image segmentation. Also, there may be two tasks for object detection, it will only have one F1-score that only considers a prediction "correct" if the model succeeds in both tasks.

In conclusion, object detection serves two fundamental tasks: classification and localization. By employing specific metrics, researchers can diagnose a model's specific flaws—whether it struggles to identify a class or fails to pinpoint an object's position. These insights will give them an option whether to restart model development, modify training parameters, or fine-tune the model further. Fine-tune the model appropriate to its application, whether precision is more important than recall, or appropriate classification from pixel-perfect detection. Metrics will be the deciding factor for determining the model's performance, and clinical usability and reliability. Ultimately, this study will utilize the metrics — sensitivity (precision and recall), specificity, F1-score, Intersection-over-Union (IoU), Area Under the Curve Receiver Operating Characteristic (AUC-ROC), prioritizing the decrease of possible false positive readings of the model, which are critical in medical imaging.

2.2 Conceptual Model of the Study

Paradigm: Input-Process-Output (IPO) with AI integration.

- Input: Chest X-ray images (preprocessed: enhancement, augmentation) from datasets like Kermany/Mendeley and gathered images from local hospitals in the Philippines.
- Process:
    - CNN backbone (YOLO) for feature extraction and classification (bacterial pneumonia/normal/other).
    - Attention modules (CBAM/SE) for focus on anomalies.
    - Grad-CAM for heatmap generation.
    - Evaluation: Precision, recall, F1, IOC, AUC-ROC; ISO 25010 for usability.
- Output: Annotated images with heatmaps and probability scores, aiding clinical decisions.

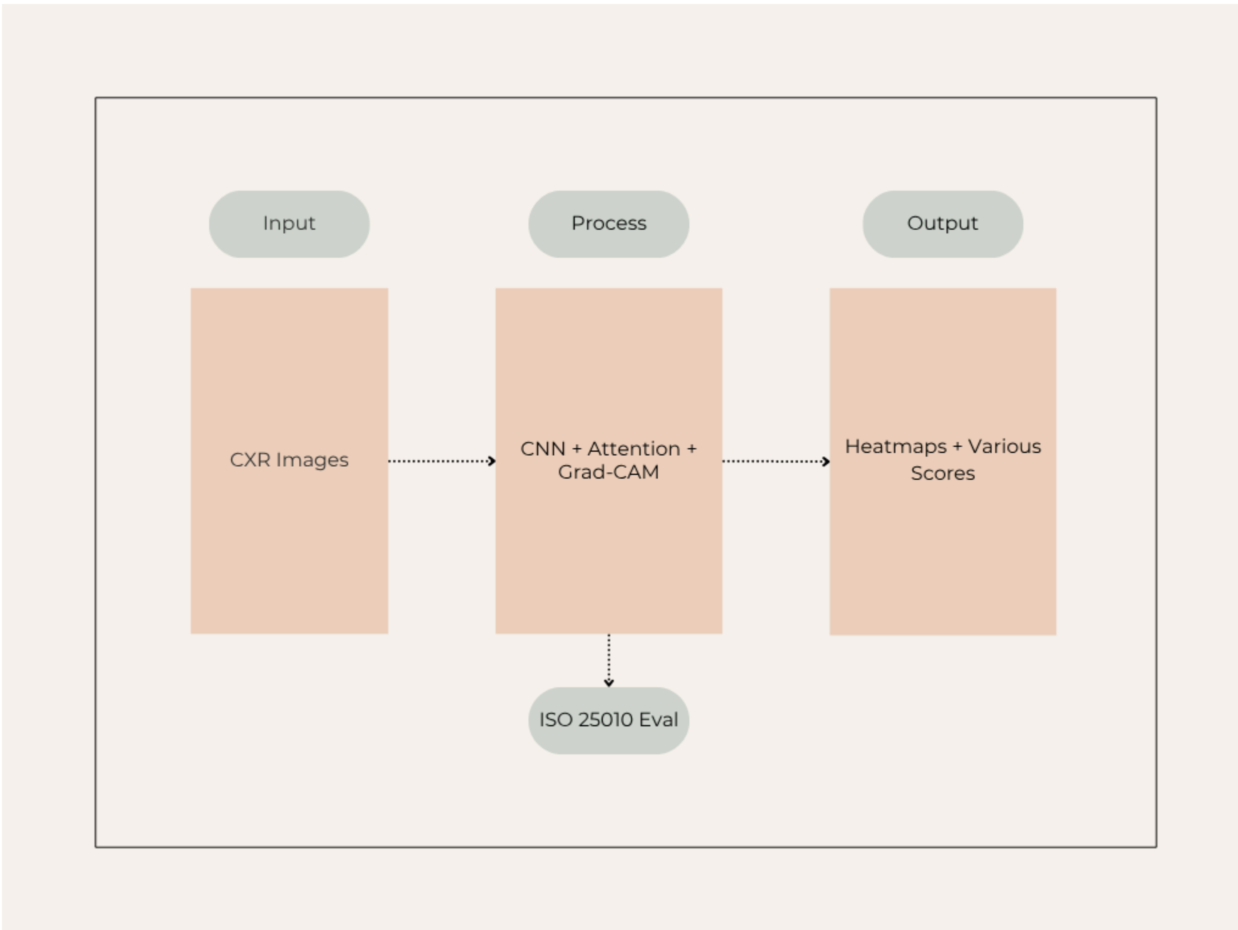

Figure 1. Conceptual Framework of RespiScan

This model synthesizes reviewed studies, ensuring explainability and performance.

2.3 Definition of Terms

- **Bacterial Pneumonia:** Lung infection by bacteria like Streptococcus pneumoniae, showing lobar consolidation on X-rays.
- **Chest X-ray (CXR):** Frontline imaging for pneumonia, revealing opacities from early congestion to late resolution.

- **Radiologic Technologists:** Healthcare professionals who operate imaging equipment, ensure minimal radiation exposure, and perform diagnostic imaging exams such as X-rays, Computer Tomography (CT), Magnetic Resonance Imaging (MRI), and mammogram scans to patients to help doctors and treat diseases.
- **Radiologists:** A physician specialized in diagnosing and treating diseases using medical imaging techniques such as X-rays, CT Scans, MRI scans and mammograms. Radiologists interpret imaging results to diagnose patients, and write patient findings and diagnosis reports.
- **Convolutional Neural Network (CNN):** Deep learning architecture extracting image features via convolutions for classification/detection.
- **Convolutional Block Attention Module (CBAM):** Integrated mechanism of Convolutional Neural Networks to emphasize important features while neglecting irrelevant ones, enhancing the model's feature extraction and performance.
- **Gradient-weighted Class Activation Mapping (Grad-CAM):** Explainable AI technique generating heatmaps of model focus areas for clinical validation.
- **AUC-ROC:** Metric measuring model discrimination (0.9+ ideal for diagnostics).
- **ISO 25010:** Standard for software quality (functional suitability, reliability, usability).
- **Localization:** Identifying the precise location of an object within an image or frame.
- **Object Detection:** Localizing pneumonia regions via bounding boxes (e.g., YOLO).
- **Attention Mechanism**: Enhances CNN focus on salient features like infiltrates.

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CHAPTER 3

Methodology

3.1 Project Design

Respiscan will be developed using the YOLOv11 framework with CBAM for better accurate detection and feature extraction of bacterial pneumonia. The main objective of the system is to detect bacteria using the YOLOv11 model. Through this system, it will assist healthcare professionals for early detection and diagnosis of the disease, which will result in improved healthcare services and overall patient respiratory health, especially in low-resource health sector areas across the Philippines. This system will also aid in early intervention of bacterial pneumonia, which is crucial in reducing the risk of further severe complications, prolonged hospitalization, and death.

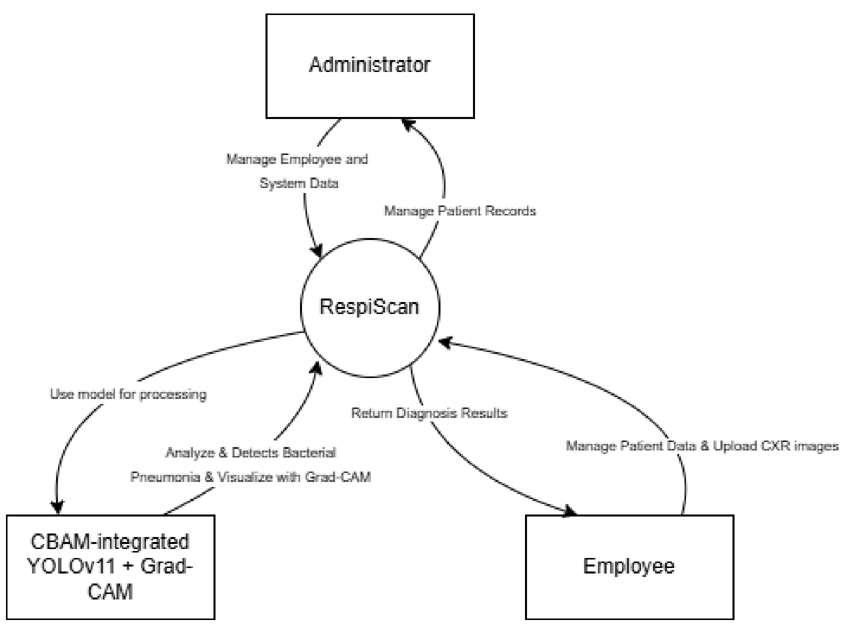

Figure 2. Context Diagram

The context diagram contains the main components that make up the system. The diagram shows the high-level system architecture, system functions, and user interaction. The system will manage patient data, system logs, and bacterial pneumonia diagnosis made by the model.

Based on the figure above, the system will initialize in system administrator, making sure that the patient data is always up-to-date. The administrator will also be responsible for employee management. Admin's role in the system is essential maintaining the system's integrity, data accuracy, and system performance.

The Employee will only be limited to managing patient health information, requesting analytics, uploading chest X-ray images for the analysis and diagnosis. After uploading the image, the RespiScan will utilize the YOLOv11 model with CBAM to analyze the model and return its corresponding diagnosis back to the employee. Results returned will then be used for further medical care and treatment.

The RespiScan containing machine learning and structured data management will improve speed and efficiency of diagnosis of bacterial pneumonia. Additionally, outputs made by the model can be utilized for improving the model's accuracy and performance in detecting the disease.

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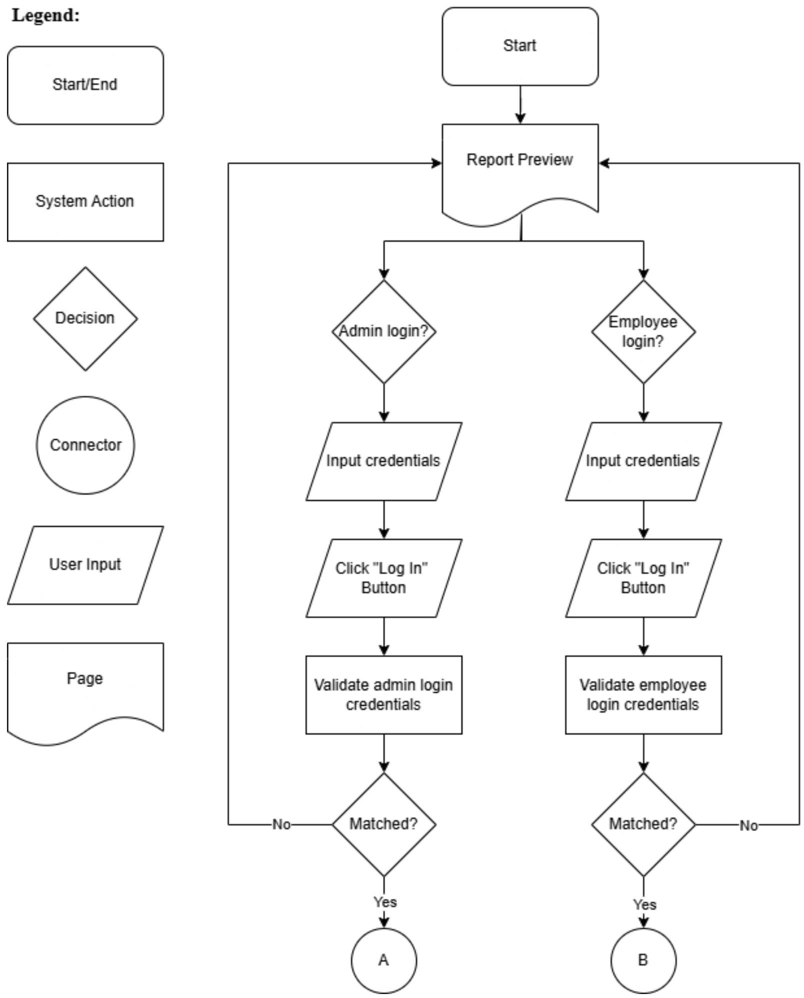

Figure 3. User Flow Diagram of Login

Figure 3 indicates the flow of interaction with the users of the system. The system will only have one login interface handling the signing in of users, both system administrators and employees. User-inputted credentials will be validated whether they have an admin or employee role. Successful login will redirect them to their corresponding interfaces. Users will be required to log in within the system before accessing its core features, which are AI diagnostics, patient and employee management, and analytics generation.

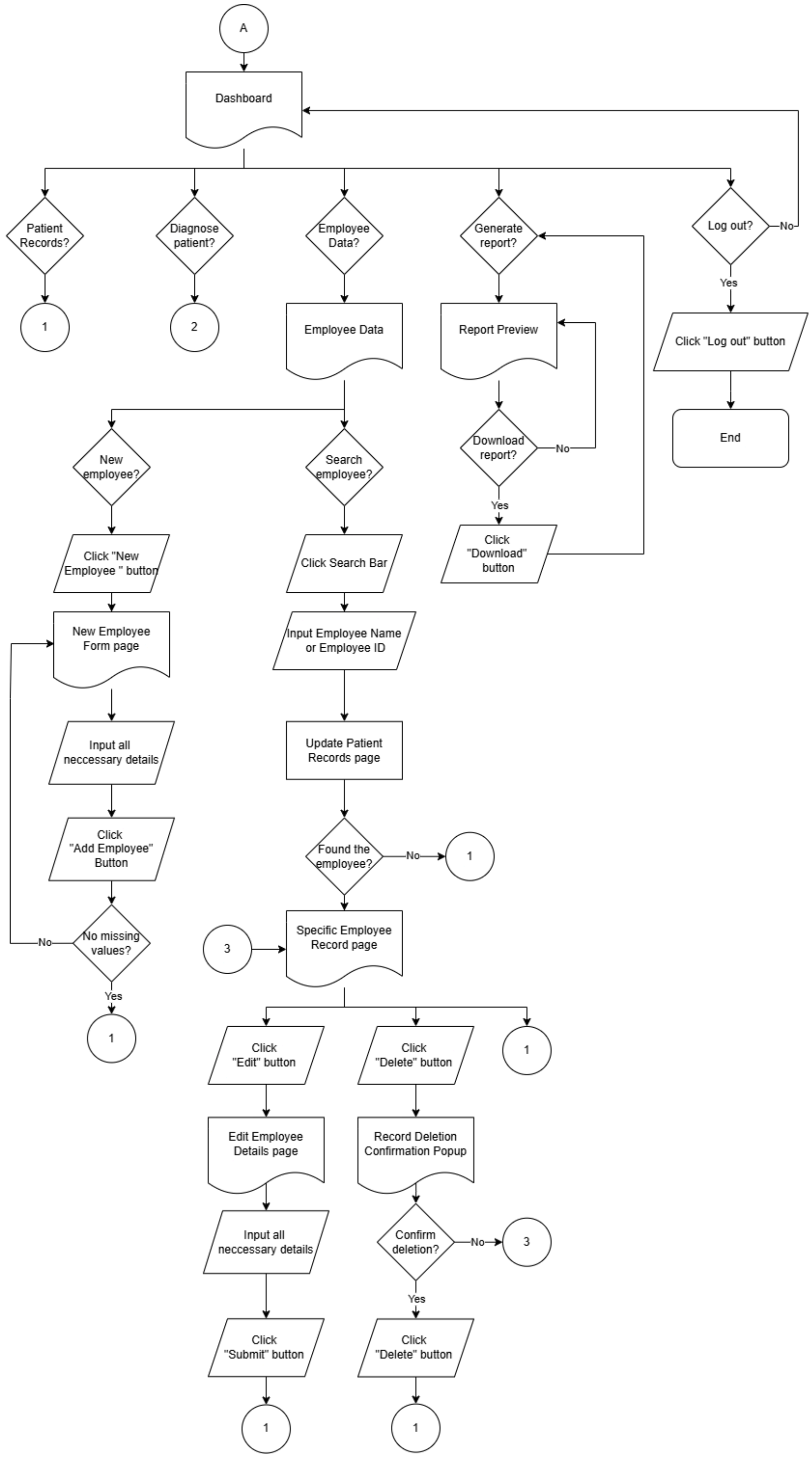

Figure 4. User Flow Diagram of Administrator

Figure 4 highlights the interaction flow for administrators after user credentials was validated and has an admin role. Admins has access to patient record management, employee management, AI-integrated diagnostics and analytics report generation. Employees can preview and download the generated analytics report by the system. The key difference between admin role and employee role is the access to employee management, which is responsible for employee accounts within the system. Only admins can create, view, modify, and delete employee data.

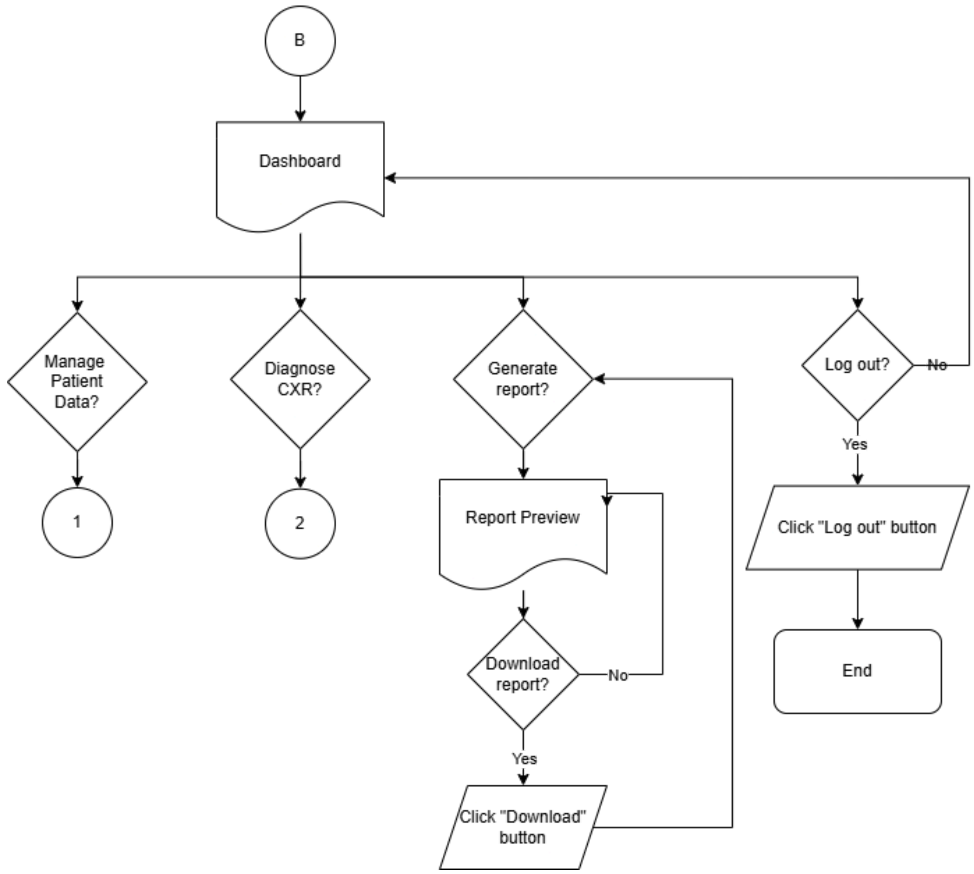

Figure 5. User Flow Diagram of Employee

Figure 5 presents the user flow diagram of the employee after it was validated from logging in the system. The user will have a dashboard, displaying the information they need on a daily basis. The employee has access to patient record management, AI-integrated diagnostics and analytics report generation. Employees can also preview and download the generated analytics report by the system.

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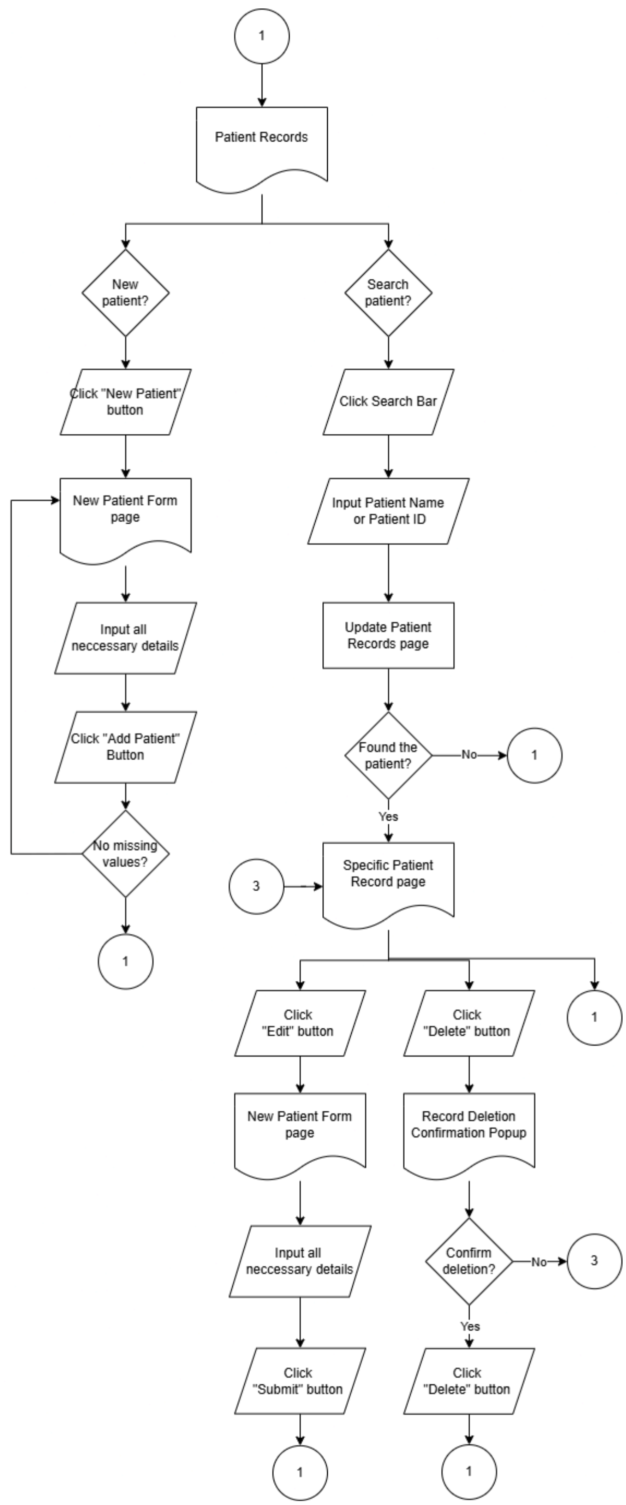

Figure 5. User Flow Diagram of Patient Records Page

The figure above demonstrates the user flow after all the users, both admins and employees, click the “Patient Records” button. This will allow them to access stored patient health records within the system. This section will have a search bar for faster retrieval of records of a specific patient. Both admins and employees can create, view, modify, and delete patient records.

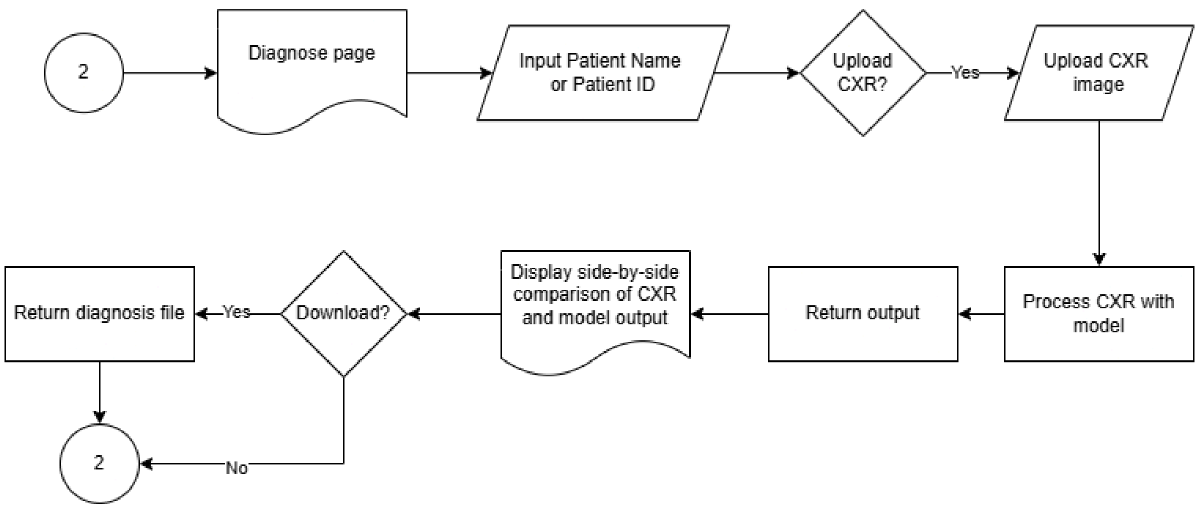

Figure 6. User Flow Diagram of Diagnosis Page

Figure 6 displays the user flow in the Diagnosis page, which is responsible for AI-driven patient diagnostics. This page is accessible by both admins and employees. The user will input the patient name or patient ID to determine whose chest x-ray image will be analyzed by the model. The user will then upload the chest x-ray image which takes time before the model returns its diagnosis regarding the image. The user will have a display of side by side comparison of initial chest x-ray image and model output with Grad-CAM visualization. The model output will also be downloadable as a file.

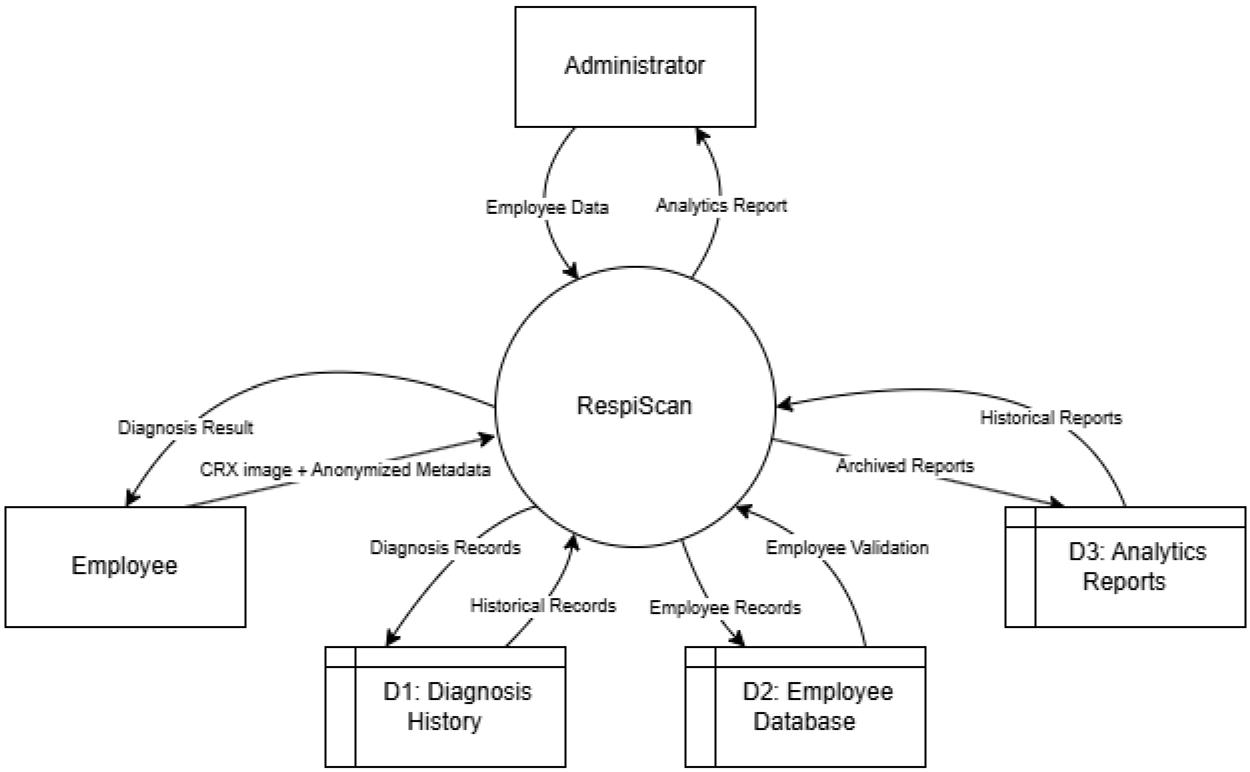

Figure 7. Data Flow Diagram Level 0

The Level 0 DFD (Figure x) illustrates the primary flow of data into and out of the RespiScan system. The Employee provides the CRX with a few anonymized metadata (age, gender). The Administrator supplies employee account information and can request a collection of reports. The System writes into two data points: Diagnosis History (D1), which holds and archives all AI-diagnostic reports, and Employee Database (D2), which holds credentials and profile information.

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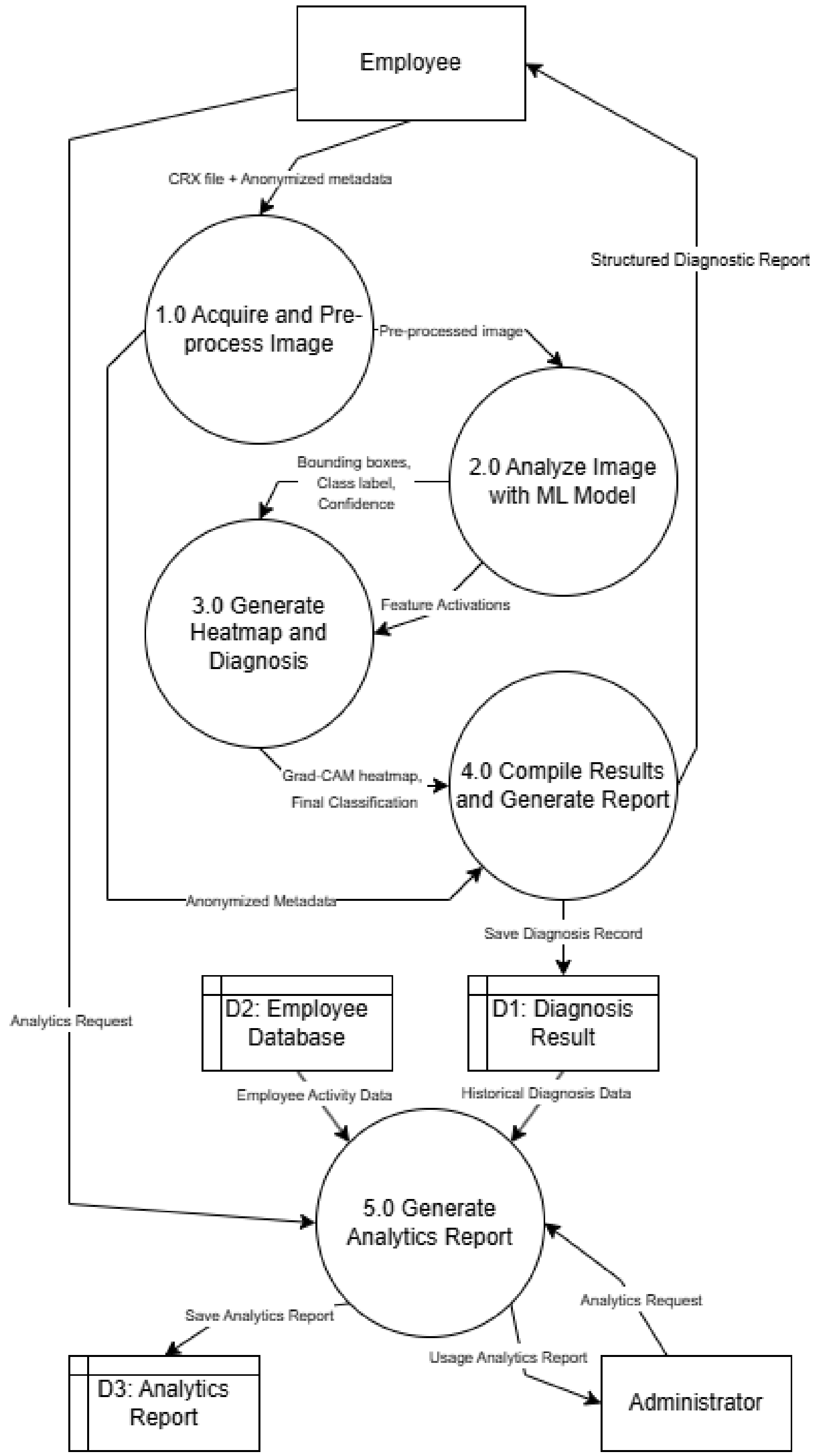

Figure 8. Data Flow Diagram Level 1

In RespiScan Data Flow Diagram Level 1, there are two main external elements: Employees and Administrator. Employees interact with the system by inputting CRX images and patient information metadata while the Administrators are responsible for maintaining patient records, diagnostic data, and analysis reports through Diagnosis Result (D1), Employee Database (D2), and Analysis Report (D3) data storage.

The diagnosis starts at Process 1.0: The employee uploads a Digital Imaging and Communications in Medicine (DICOM) or a standard image file. The image is resized to a uniform dimension (e.g., 720x720 pixels), normalize pixel intensities, and apply contrast-limited adaptive histogram equalization (CLAHE) to enhance subtle opacities. No identifiable information about the patient is passed down, only anonymized information like age and gender to help in diagnosis.

After the pre-processing of the image, we move on to Process 2.0: Analyze image with ML model. The image is passed onto the YOLOv11 backbone. The CBAM module re‑weights feature maps along both channel and spatial axes, forcing the network to focus on salient, pneumonia‑indicative patterns (consolidations, air bronchograms). The model outputs bounding box coordinates, a class label, and a confidence score for each detected region of interest.

Process 3.0: After the feature activation in the CNN, Grad-CAM computes a localization heatmap that highlights the areas of the images that is most qualified for the “Bacterial Pneumonia” classification. The heat-mapped image is then presented side by side with the original CXR, producing an interpretable visualization. The final classification as well as the confidence score are recorded alongside the heatmap

The findings of the past processes are then given to Process 4.0 Compile and Generate Report: Result from the previous steps are compiled and used to construct a structured diagnosis report. This report includes the patients anonymized information, the predicted label confidence score, and the Grad-CAM heatmap as a supplementary image. The report is then stored in the Diagnosis History data store for future reference and analytics, and simultaneously be available for download by the employee.

Lastly, Process 5.0: Generate Analytics Report: Both the Administrator and Employees can request analytics reports. The administrator can request system statistics and audit logs, while employees can request ranged historical analytics, such as weekly/monthly averages.

Throughout the processes, the system is controlling access based on the user's role (Admin or Employee), and all interactions with stored data are audit logged to ensure no abuse of data is being committed.

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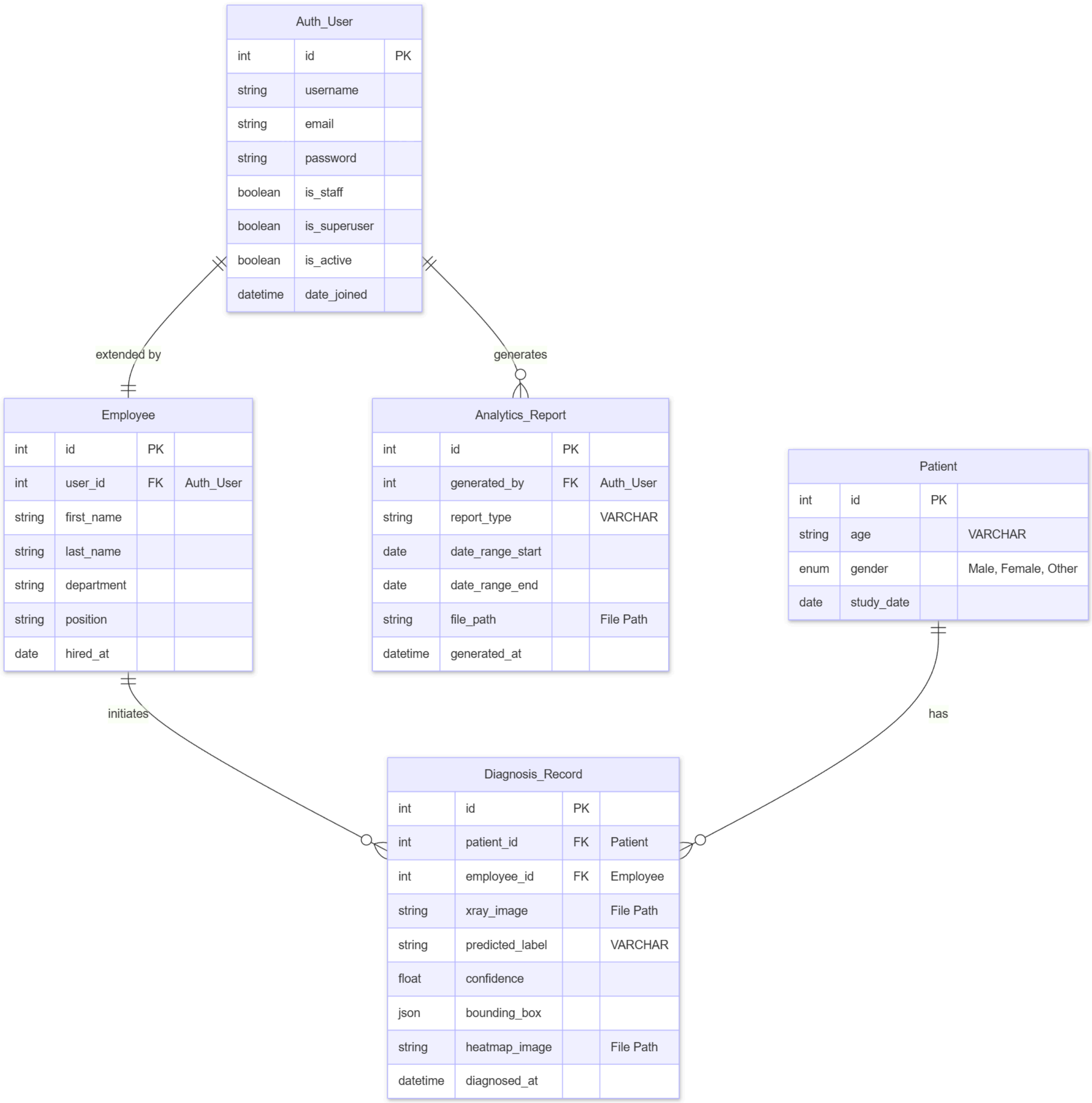

Figure 9. Entity Relationship Diagram (ERD)

The ERD describes essential data entities and their respective relationships. It is made minimal to protect patient privacy while still supporting model evaluation and clinical audit trials.

The relationships in the diagram are straightforward: One employee can initiate many Diagnosis_Records; one Patient can have multiple Diagnosis_Records (e.g., follow-up CXRs); each Diagnosis_Report corresponds to exactly one Patient and one Employee. The Administrator (Auth_User) can generate multiple Analytics_Reports. Together, these artifacts create the blueprint for RespiScan, which separates user authentication, staff data, patient data, and AI-generated diagnostic results, facilitating both clinical review and future model retraining.

3.2 Project Development

Project Development presents the detailed process of developing RespiScan: An AI-Powered Software using CNN-Based AI Models in Bacterial Pneumonia in Chest X-rays using YOLOv11 integrated with CBAM..

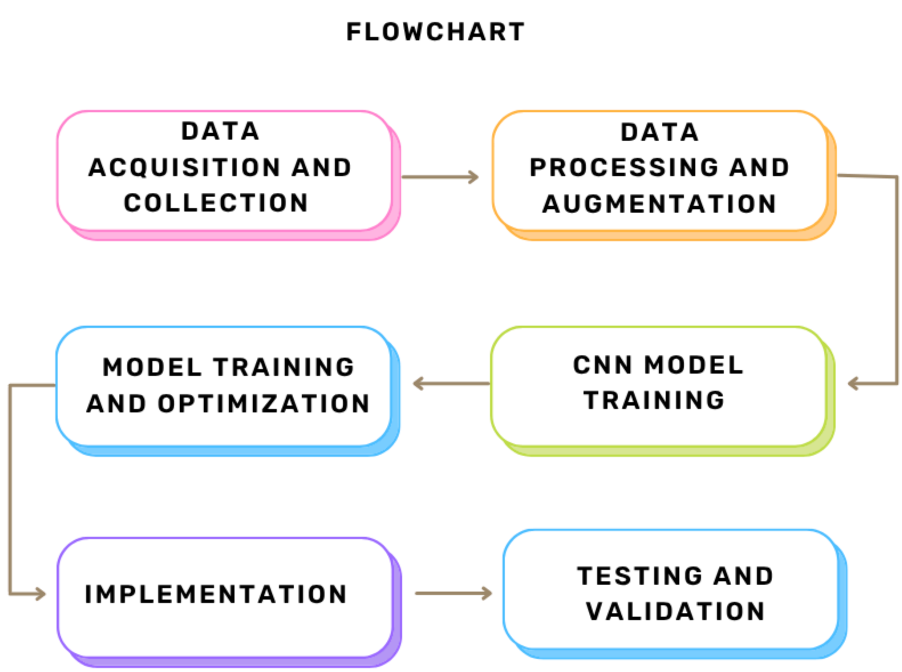

Figure 10. Flowchart of Project Development Phases

1. Data Acquisition and Collection

The initial stage of building the RespiScan software focuses on collecting a large and medically valid collection of chest X-ray pictures. To train the Convolutional Neural Network (CNN) efficiently, the dataset must have a fair representation of two basic classifications: normal (healthy) lungs and lungs afflicted with bacterial pneumonia. These pictures will be obtained from verified, publically available medical sources (such as Kaggle, Mendeley Data, or the NIH Chest X-ray dataset) and acquired datasets from local hospitals to guarantee that the model learns from a diverse range of anatomical structures and infection severity.

2. Data Processing and Augmentation

Because raw medical images can vary in size, resolution, and quality, they must be standardized before being fed into an AI model. In this step, all obtained chest X-rays will be preprocessed. This includes resizing the images to the uniform dimensions required by the specific CNN architecture, as well as normalizing pixel values to ensure consistent brightness and contrast. Additionally, data augmentation techniques such as small rotations, zooming, and horizontal flipping will be used. This artificial enlargement of the dataset prevents the model from overfitting and increases its capacity to generalize when interpreting previously unknown X-rays.

3. CNN Model Training

To initiate the deep learning process, the processed chest X-ray dataset is divided into training and validation sets. The chosen CNN architecture is configured with focus techniques (such as CBAM), and training data is input into the network. The model's convolutional layers will learn to extract essential spatial data and visual patterns that differentiate bacterial pneumonia infiltrates from healthy lung tissue over several iterations (epochs). Throughout this process, the model constantly modifies its internal weights to reduce classification mistakes and enhance diagnostic accuracy, establishing a course for the eventual development of Grad-CAM heatmaps.

4. Model Training and Optimization

Architecture Selection: The team utilizes CNN-based models, which are the industry standard for identifying spatial hierarchies in images.

Training Process: The model is trained over multiple epochs. Success is measured using metrics such as:

- **Accuracy:** Overall percentage of correct predictions.
- **Sensitivity (Recall):** The ability to correctly identify all actual pneumonia cases (crucial in medical settings).
- **Loss Function:** Minimized using optimization algorithms like Adam to refine the model's weight parameters.

5. Implementation (Coding Phase)

Development is executed in iterative sprints:

Iteration 1: Development of the basic UI and file upload functionality.

Iteration 2: Integration of the pre-trained CNN model into the backend.

Iteration 3: Connecting the frontend and backend via API to provide real-time classification results.

6. Testing and Validation

To ensure the system is reliable for healthcare use, it undergoes:

Model Validation: Testing the AI on a "hidden" dataset it hasn't seen before to check for overfitting.

User Acceptance Testing (UAT): Reviewing the software's usability to ensure it fits the workflow of a clinical environment.

3.3 Testing and Operation Procedure

Table 1. Admin Interface

System Function	Procedure	Expected output
1. Login as Admin	1. Input username 2. Input password 3. Click the login button	1. The system must check if the username and password is valid or invalid without crashing. 2. Incorrect credentials should prompt an error message saying that the credentials are not correct 3. If logged in successfully, the user will be redirected to the admin dashboard.
2. Download Analytics	1. Navigate to the dashboard 2. Download the	1. The system generates a downloadable analytics report.

	Analytics report by clicking the "Generate & Print" Button	
3. Manage Employees	1. Click "Employee" tab 2. Add Employee by clicking "Add" Button 3. Fill out necessary details. 4. Edit Details by clicking the edit icon. 5. Delete an employee by clicking the delete icon 6. Save changes by clicking the "update button"	1. Employee accounts are updated accordingly 2. Error messages appear for missing or invalid inputs.
4. AI Diagnosis	1. Click the "Diagnose" Icon. 2. Upload an CXR image to be analyzed by the model 3. Click " Analyze CXR" icon and then the diagnosis will be shown. 4. After the results, you may download the generated report of the patient by clicking the "View Report" button.	1. The System analyzes the image and display the diagnosis results. 2. The system generates a downloadable report of the diagnosis.
6. Logout	1. Click the "Logout" button.	1. The system logs out the admin and redirects to the login screen.

Table 2. Employee Interface

System Function	Procedure	Expected Output
1. Login Interface	1. Input Username 2. Input Password 3. Click Sign in Button	1. The system must accept the valid and invalid username password without crashing. 2. Incorrect credentials should prompt an error message saying that the credentials are incorrect. 3. Once logged in successfully, the user will be redirected to the employee dashboard.
2. Generate Report	1. Navigate through the dashboard 2. Click the “Generate Report” button.	1. The system will generate a downloadable analytics report.
3. Manage Patients	1. Navigate through the dashboard and click the “Patient Records” button. 2. Click “Add Patient” Tab 3. Fill out the necessary details. 4. Search patients by searching on the search bar. 5. Edit Details by clicking the edit icon 6. Delete a patient by clicking the “update” button	1. Patient Information is updated accordingly. 2. Error messages appear for missing or invalid inputs.

5. AI Diagnosis	1. Click the "Diagnose" Icon of the chosen patient. 2. Upload an CXR image to be analyzed 3. Click the "Analyze" Icon and then the diagnosis will be shown 4. After the results, you may download the generated report of the patient by clicking the "View Report" icon.	1. The system analyzes the image and displays the diagnosis results. 2. The system generates a downloadable report of the diagnosis.
6. Logout	1. Click the "Logout" button.	1. The system logs out the user and redirects to the login screen.

3.4 Evaluation Procedure

This section will describe how RespiScan will be evaluated to ensure that it meets its objectives as an AI-powered Bacterial Pneumonia detection system. The evaluation will involve projected end-users and technical experts to assess the system's usability, functionality, performance, reliability, and security in accordance with ISO/IEC 25010 software quality standards:

Admin End-user Evaluation

The administrative evaluation focuses on the management, security, and maintenance of the platform. Administrators will assess the system's Security through user access controls and the Maintainability of the database and CNN model updates. This ensures that the back-end infrastructure is robust enough to handle sensitive medical data while remaining easy to manage.

Patients End-user Evaluation

This evaluation measures the transparency and clarity of the AI-generated results from a non-clinical perspective. Patients will evaluate the system's Usability, specifically focusing on whether heatmaps and reports make the diagnosis understandable and trustworthy.

Medical Professionals End-user Evaluation

Pulmonologist and radiologist will provide the most critical feedback regarding the software's Functional Suitability and clinical accuracy. They will assess the Performance Efficiency by measuring how quickly the AI generates results and the clarity of the readings identifying pathological regions. Their input ensures that RespiScan functions as a reliable diagnostic aid that aligns with established medical ground truths.

Experts End-user Evaluation

This group, consisting of IT and Software Engineering professionals, will evaluate the technical integrity and architecture of the system. They will focus on Reliability and Compatibility, testing how the software handles edge cases, corrupted data, and cross-platform deployment. This technical audit ensures the software meets the "Maturity" and "Fault Tolerance" standards required by the ISO/IEC 25010 framework.

Evaluation Criteria and Tools

The evaluation will use ISO/IEC 25010-based criteria such as usability, accuracy, reliability, performance efficiency, and security. Data will be collected using Likert-scale questionnaires and observation checklists to ensure a consistent, measurable, and comprehensive assessment.